

Software Requirement Specification

COS 301 - Capstone 2026
Capstone Project: EquityLens
By The Big Five



Project Name: EquityLens
Client: Amazon Web Services (AWS)
Team: The Big Five (TB5)

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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for EquityLens, an investment analytics platform. It is intended for the development team (The Big Five - TB5), project examiners and the stakeholders at AWS to define the scope and success criteria for the project. It is written so that a developer with no prior knowledge of the codebase could implement the system from the requirements without being constrained to specific technologies.

1.2 Business Need

Retail investing in South Africa has surged due to accessible brokerage platforms, but a significant analytical gap remains, this gap entails retail investors having access to the market yet lack the tools needed to interpret risk metrics and connect world events to portfolio volatility. Investors face data overwhelm - they hold diversified portfolios but cannot easily calculate total risk exposure, or understand why specific stocks moved, without manual and error-prone spreadsheets.

EquityLens addresses this by centralising investment data into a single dashboard. It automates the calculation of financial indicators, correlates news events with price movements, and uses a conversational AI assistant to explain metrics in plain language - bridging the gap between raw data and actionable insight, at no cost to the user.

1.3 Scope

EquityLens is a web-based analytics suite for South African retail investors. It covers:

- Secure user registration, email verification, multi-factor authentication and login
- Portfolio ingestion via spreadsheet upload (CSV/Excel), brokerage PDF statement upload, or manual entry
- Automated financial indicator calculation with plain-language explanations
- News-to-price-movement correlation for the user's holdings
- A conversational AI assistant with persistent conversation history and safety guardrails
- A dashboard summarising portfolio value, health, performance against a benchmark, and recommended actions
- A watchlist for stocks the user does not yet own

Out of scope: direct stock trading, brokerage integration that requires the user's brokerage credentials, and regulated financial advice.

1.4 Definitions, Acronyms and Abbreviations

- EquityLens - the investment analytics web platform described in this document
- SRS - Software Requirements Specification

- SAS - Software Architecture Specification
- FR - Functional Requirement
- NFR - Non-Functional Requirement
- UC - Use Case
- US - User Story
- TUCBW - This Use Case Begins When
- TUCEW - This Use Case Ends When
- MFA - Multi-Factor Authentication
- TOTP - Time-based One-Time Password
- API - Application Programming Interface
- AWS - Amazon Web Services
- POPIA - Protection of Personal Information Act (South Africa)
- JSE - Johannesburg Stock Exchange
- ALSI - All Share Index (JSE)
- LLM - Large Language Model
- ETF - Exchange Traded Fund
- CAPM - Capital Asset Pricing Model
- RSI - Relative Strength Index
- P/E - Price-to-Earnings ratio
- CI/CD - Continuous Integration / Continuous Deployment

1.5 Document Overview

Section 2 describes the intended users and their goals as user stories. Section 3 presents the use case model: a use case diagram followed by a TUCBW/TUCEW description of each use case. Section 4 specifies the functional requirements, assigned to eight subsystems. Section 5 quantifies the five targeted quality requirements. Section 6 presents the domain model as a UML class diagram with entity descriptions. Section 7 traces each use case to the functional requirements that realise it, and Section 8 records the assumptions and external dependencies the requirements rest on. How the system is structured to satisfy these requirements is specified separately in the Software Architecture Specification (SAS).

2. User Stories and User Characteristics

2.1 Intended Users

Retail Investor - the primary and only in-app user. A South African retail investor who imports portfolio data and uses the analytics, news, AI assistant and watchlist features. Assumed to have basic familiarity with stock tickers but no formal finance training therefore everything must be explainable in plain language.

EquityLens deliberately has a single in-app role. System administration (account management, configuration, monitoring) is an operational responsibility carried out by the development team through the hosting platform's own tools and it is not a user-facing capability of the system and no administrative interface is in scope.

2.2 User Stories

2.2.1 Authentication

- US-1: As a new user, I want to create an account so that my portfolio data is saved securely.
- US-2: As a new user, I want to verify my email address with a code so that only I can activate my account.
- US-3: As a security-conscious user, I want to protect my account with an authenticator app code so that a stolen password alone cannot access my data.
- US-4: As a returning user, I want to log in so that my session remains active while I work.
- US-5: As a user on a shared device, I want to log out so that others cannot access my data.

2.2.2 Portfolio Management

- US-6: As an investor, I want to upload a spreadsheet of my holdings so that I don't have to enter them manually.
- US-7: As an EasyEquities user, I want to upload my PDF statement so the system can extract my holdings automatically.
- US-8: As a user with a small portfolio, I want to manually add a holding so I can track new investments immediately.
- US-9: As a cautious user, I want to preview what the system extracted from my file before it is saved so that mistakes don't corrupt my portfolio.

2.2.3 Analytics and Financial Indicators

- US-10: As an investor, I want to see my portfolio's Sharpe Ratio so that I can judge whether my returns justify the risk.
- US-11: As a novice, I want a plain-language explanation next to every indicator so I can learn what the numbers mean.
- US-12: As a technical user, I want to see the RSI for my stocks so I can identify overbought or oversold conditions.

2.2.4 AI Assistant

- US-15: As a novice, I want to ask “Is my portfolio diversified?” and get a natural-language answer grounded in my actual holdings.
- US-16: As a returning user, I want my previous AI conversations saved so I can pick up where I left off.
- US-17: As a user, I want to rename or delete old conversations so my chat history stays organised.

2.2.5 Dashboard and Insights

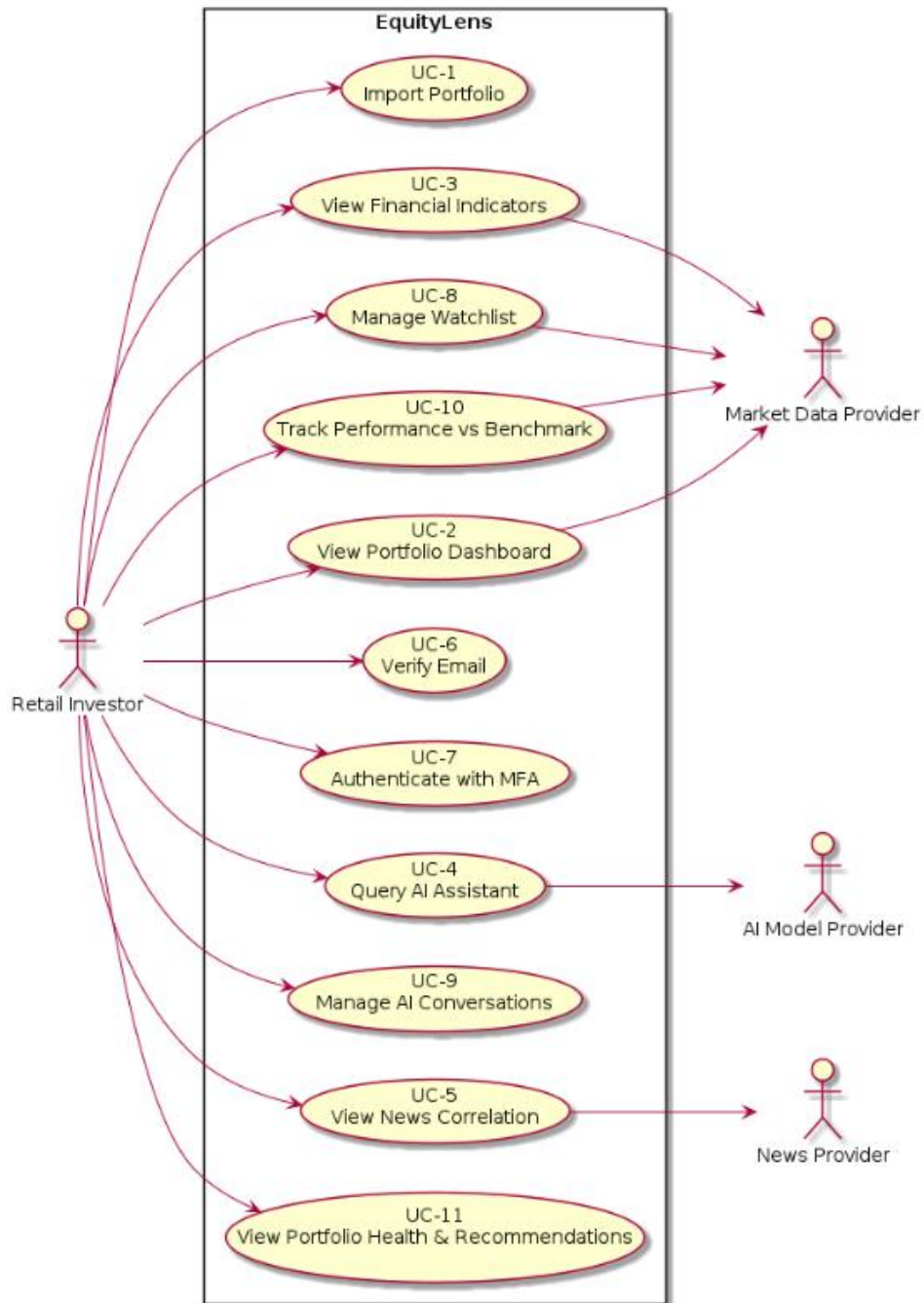
- US-18: As an investor, I want a single health score for my portfolio so I can tell at a glance whether anything needs attention.
- US-19: As an investor, I want to see my performance charted against the JSE All Share Index so I know if I’m beating the market.
- US-20: As a busy user, I want the system to tell me what needs my attention (concentration risk, poor diversification) rather than making me hunt through data.
- US-21: As an investor, I want to see which holdings drove today’s gain or loss so I know why my portfolio moved.

2.2.6 Watchlist

- US-22: As an investor, I want to add stocks I don’t yet own to a watchlist so I can track them before buying.
- US-23: As a user, I want to remove a stock from my watchlist when I’m no longer interested.

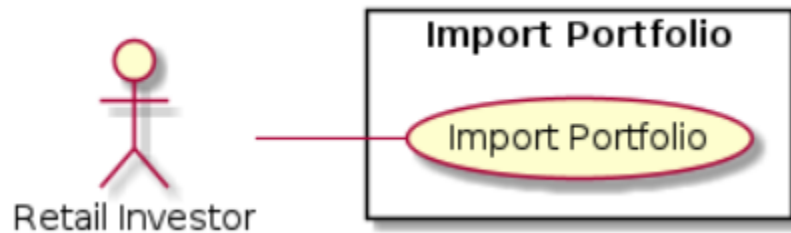
3. Use Cases and Use Case Diagram

3.1 Use Case Descriptions



3.2 Use Case Descriptions

UC-1: Import Portfolio



Actor: Retail Investor

Precondition: The user is authenticated and on the Portfolio page. The user has a portfolio file (CSV, Excel) or details needed for manual entry.

Postcondition: The user's holdings are saved against their account and visible in the holdings table.

TUCBW the user clicks the Import button on the Portfolio page.

Main flow:

1. The system displays the upload window.
2. The user selects a CSV file from their device.
3. The system extracts the holdings from the file.
4. The system validates the extracted content.
5. The system displays a preview of the parsed holdings for confirmation.
6. The user confirms the preview.
7. The system saves the holdings and displays them in the holdings table.

Alternative flows:

- Excel upload: at step 2 the user selects an Excel workbook that follows the provided template; the flow continues from step 3.
- Manual entry (fallback): at step 1 the user opens the manual entry form instead, captures ticker, quantity and purchase price, and the flow joins at step 4.

Exception flows:

- Malformed file: validation at step 4 fails. The system rejects the upload, shows an error message, and keeps the upload window open so the user can retry.
- No content found: extraction succeeds but no holdings are found. The system informs the user and prompts them to enter holdings manually or upload a different file.

TUCEW the user sees their imported holdings displayed in the holdings table.

UC-2: View Portfolio Dashboard



Actor: Retail Investor

Precondition: The user is authenticated.

Postcondition: The dashboard reflects the user's current portfolio state.

TUCBW the user navigates to the Dashboard page.

Main flow:

1. The system confirms the user's session is valid.
2. The system retrieves the user's holdings, current market values, sector breakdown, performance history and dividend history.
3. The system displays the portfolio value, today's change, the largest positions, sector allocation, and the performance chart.

Alternative flow:

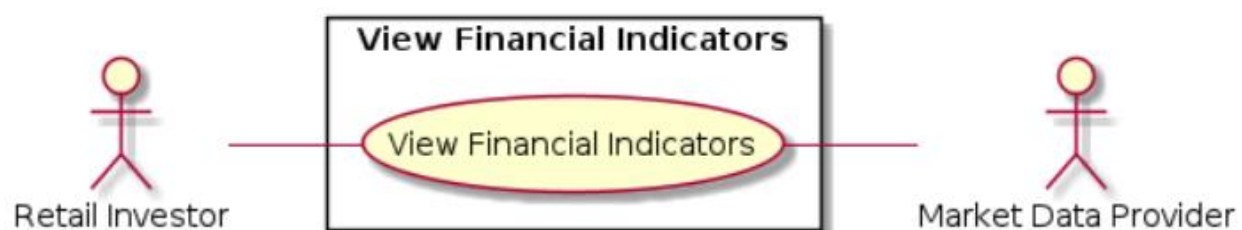
- Empty portfolio: at step 2 no holdings exist. The dashboard displays an onboarding state prompting the user to import a portfolio, with a direct link to the Portfolio page.

Exception flows:

- Unauthenticated user: the session is invalid or missing; the system redirects the user to the login page and the dashboard is not displayed.
- Data retrieval failure: at step 2 the system cannot load the portfolio; the dashboard displays an error state with a retry prompt.

TUCEW the user sees their populated dashboard (or the onboarding state for an empty portfolio).

UC-3: View Financial Indicators



Actor: Retail Investor

Precondition: The user is authenticated and has at least one holding. Sufficient price

history exists to compute each indicator.

Postcondition: An indicator card is displayed for each of the seven supported indicators.

TUCBW the user navigates to the Analytics page.

Main flow:

1. The system retrieves the user's saved holdings.
2. The system calculates the seven supported indicators (Sharpe Ratio, Sortino Ratio, P/E Ratio, CAPM expected return, Altman Z-Score, Beta, RSI) for the relevant holdings.
3. The system displays one card per indicator showing its name, value, and classification where applicable.

Alternative flow:

- View explanation: the user hovers over (or focuses) an indicator name; the system displays a plain-language explanation. Moving away hides it without affecting the page.

Exception flows:

- Insufficient data: an indicator cannot be calculated for a holding. Its card shows an N/A state with a short explanation; all other cards render normally.
- Calculation failure: a calculation fails. The affected card shows an error state with a retry prompt; the rest of the page remains usable.

TUCEW the user sees the indicator cards for their portfolio.

UC-4: Query AI Assistant



Actor: Retail Investor

Precondition: The user is authenticated and on the AI Assistant page.

Postcondition: The assistant's response is displayed and the exchange is stored in the active conversation.

TUCBW the user submits a typed question in the chat input field.

Main flow:

1. The system sends the question, together with the user's portfolio context, to the AI layer.

2. The AI layer generates a response grounded in the user's holdings, indicators and relevant market data.
3. The system displays the response beneath the user's message, with a disclaimer that responses are informational only.

Alternative flow:

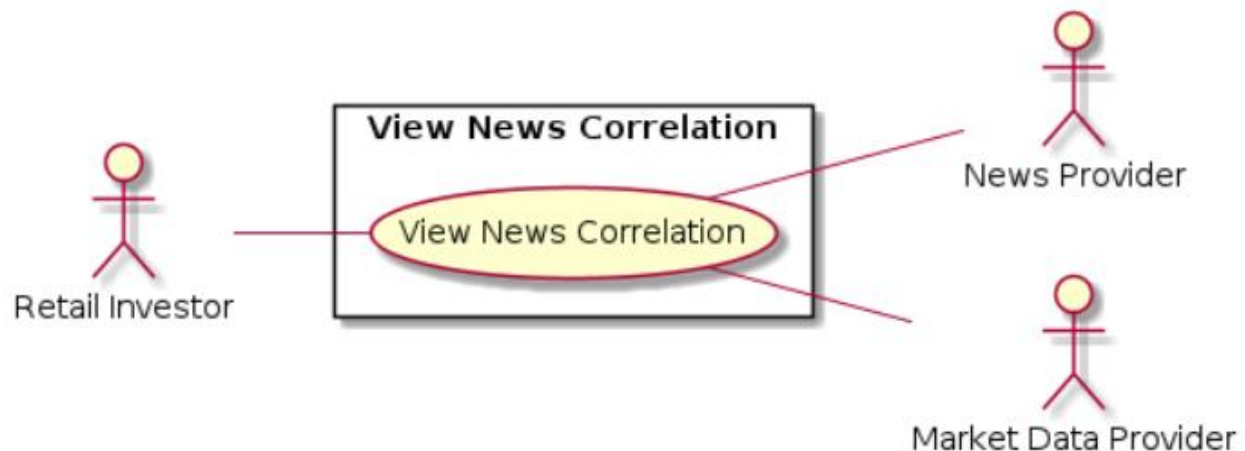
- Suggested prompt: instead of typing, the user clicks one of the suggested prompt chips shown on load; the chip submits that question and the flow continues from step 1.

Exception flows:

- Guardrail triggered: the question asks for an explicit buy or sell recommendation. The system responds that EquityLens provides analysis and information only, and directs the user to the Analytics page to inform their own decision.
- AI service unavailable: the primary AI layer fails. The system falls back to the secondary layer automatically. If both are unavailable, an error message prompts the user to try again.

TUCEW the user sees the assistant's response in the conversation.

UC-5: View News Correlation



Actor: Retail Investor

Precondition: The user is authenticated and on the News page.

Postcondition: News articles relevant to the user's holdings are displayed, filterable by ticker and date.

TUCBW the user navigates to the News page.

Main flow:

1. The system loads articles relevant to the user's holdings.

2. The system displays each article with headline, source, publication time, related ticker badge, and a price-movement indicator showing how the related stock moved around publication.
3. The user clicks an article; the full article opens in a new tab at the source publication.

Alternative flows:

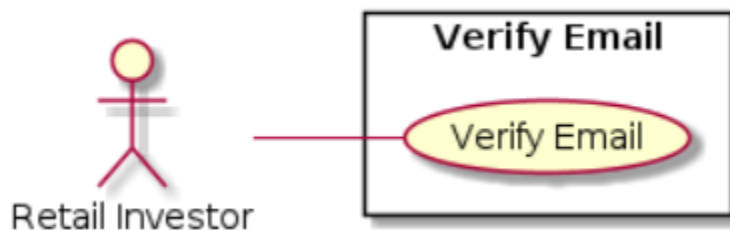
- Filter by ticker: the user clicks a ticker chip; the list filters to articles correlated with that holding.
- Filter by date range: the user selects Today, This Week, This Month, or a custom range; the list filters to that window.

Exception flows:

- No articles found: an empty state prompts the user to broaden the filter or check back later.
- News service unavailable: an error message indicates news is temporarily unavailable and prompts the user to try again.

TUCEW the user sees the news list relevant to their filters.

UC-6: Verify Email



Actor: Retail Investor

Precondition: The user has just registered an account and received a 6-digit verification code by email.

Postcondition: The account is activated and the user can log in.

TUCBW the user opens the email verification page after registering.

Main flow:

1. The system displays the verification form, pre-filled with the user's email address where available.
2. The user enters the 6-digit code from their email.
3. The user submits the code.
4. The system confirms the code is correct and activates the account.
5. The system displays a success message and redirects the user to the login page.

Alternative flow:

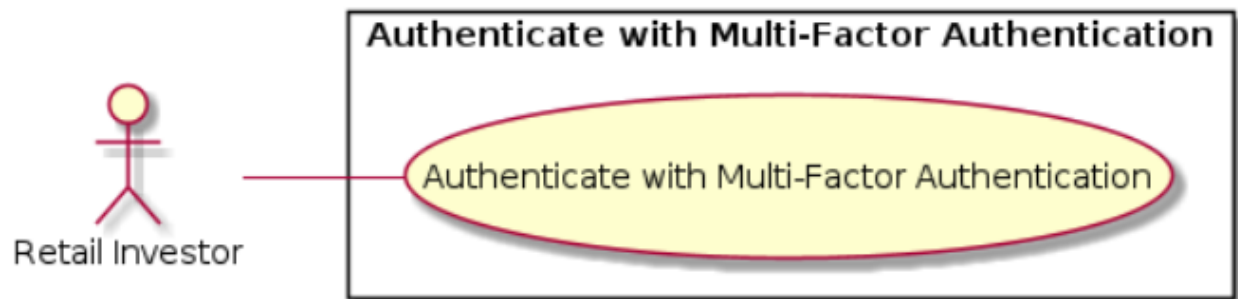
- Arriving without a pre-filled email: the user reaches the page directly; the system additionally asks for the email address before the code can be submitted.

Exception flow:

- Invalid or expired code: the system rejects the code, displays an error message, clears the code field, and allows the user to retry.

TUCEW the user sees the success confirmation and is redirected to the login page.

UC-7: Authenticate with Multi-Factor Authentication



Actor: Retail Investor

Precondition: The user has a verified account and is on the login page.

Postcondition: The user is authenticated with a second factor and lands on the dashboard.

TUCBW the user submits a correct email and password on the login page.

Main flow (returning user with MFA configured):

1. The system prompts for a 6-digit code from the user's authenticator app.
2. The user enters the code and submits.
3. The system verifies the code and completes authentication.
4. The system redirects the user to the dashboard.

Alternative flow (first login - MFA setup):

1. The system determines that MFA has not yet been configured and displays a QR code.
2. The user scans the QR code with an authenticator app (or reveals and enters the setup key manually).
3. The user enters the 6-digit code the app generates, to prove the pairing works.
4. The system activates MFA for the account, completes authentication and redirects the user to the dashboard.

Exception flow:

- Invalid code: the system rejects the code, displays an error message, clears the input and allows the user to retry.

TUCEW the user lands on their dashboard, authenticated with both factors.

UC-8: Manage Watchlist



Actor: Retail Investor

Precondition: The user is authenticated.

Postcondition: The user's watchlist reflects the additions or removals made.

TUCBW the user opens the watchlist and chooses to add or remove a stock.

Main flow (add):

1. The user searches for a stock by ticker or company name.
2. The system displays matching stocks.
3. The user selects a stock to add.
4. The system saves it to the user's watchlist and displays it with its current price and daily percentage change.

Alternative flow (remove):

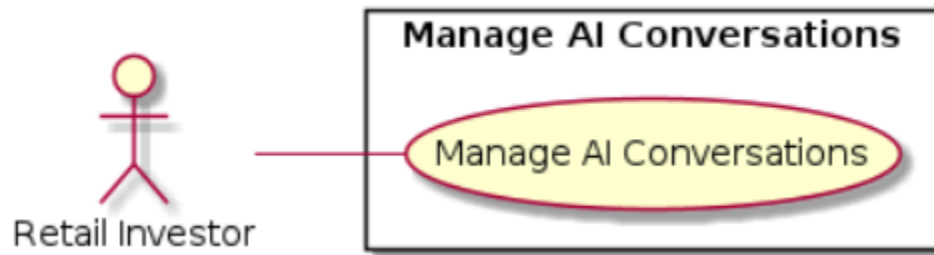
1. The user selects the remove action on a watchlist entry.
2. The system removes the entry and updates the displayed list.

Exception flows:

- No match found: the search returns no stocks; the system shows an empty result state and the user may refine the search.
- Price unavailable: a watchlist entry's market data cannot be fetched; the entry renders with its last known data or a clear unavailable state, without breaking the rest of the list.

TUCEW the user sees their updated watchlist.

UC-9: Manage AI Conversations



Actor: Retail Investor

Precondition: The user is authenticated and on the AI Assistant page.

Postcondition: The user's conversation list reflects the changes made, and the selected conversation's history is displayed.

TUCBW the user interacts with the conversation list in the AI Assistant sidebar.

Main flow (resume a conversation):

1. The system displays the user's saved conversations.
2. The user selects a conversation.
3. The system loads and displays that conversation's full message history.
4. The user continues the conversation from where it left off.

Alternative flows:

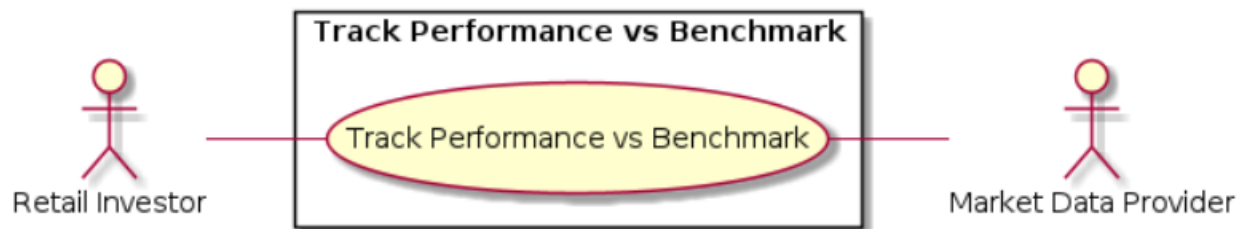
- New conversation: the user clicks New Chat; the system clears the message area and the next question starts a new saved conversation.
- Rename: the user chooses rename on a conversation, types a new title and confirms; the system saves and displays the new title.
- Delete: the user chooses delete on a conversation; the system removes it from the list, and if it was open, clears the message area.

Exception flow:

- History unavailable: the conversation list or messages cannot be loaded; the system shows the page in its empty state and the user can still start a new conversation.

TUCEW the user sees the conversation list and message area reflecting their action.

UC-10: Track Performance vs Benchmark



Actor: Retail Investor

Precondition: The user is authenticated and has holdings. At least one daily portfolio value snapshot exists.

Postcondition: The user has seen their portfolio's return charted against the JSE All Share Index.

TUCBW the user views the performance chart on the Dashboard.

Main flow:

1. The system records a snapshot of the portfolio's total value and the benchmark level for the current day (at most one per day).
2. The system retrieves the accumulated snapshot history.
3. The system displays the portfolio value and benchmark as two series on one chart, together with the portfolio return, benchmark return, and the difference over the visible window.

Alternative flow:

- Missing benchmark data: the benchmark level could not be fetched on a given day; the chart shows a gap in the benchmark series for that day rather than fabricating a value.

Exception flow:

- No history yet: fewer than two snapshots exist; the chart area explains that performance history will appear once the portfolio has accumulated a few days of data.

TUCEW the user sees their performance plotted against the benchmark with the comparative return figures.

UC-11: View Portfolio Health and Recommendations



Actor: Retail Investor

Precondition: The user is authenticated.

Postcondition: The user has seen their portfolio health score and any recommended actions.

TUCBW the user views the health score and Action Centre on the Dashboard.

Main flow:

1. The system computes a health score out of 10 from the user's actual holdings, based on diversification, concentration risk, today's momentum, and performance relative to the benchmark.
2. The system displays the score, its classification (Excellent / Healthy / Mixed / Needs attention), and the strongest and weakest contributing factors in plain language.
3. The system evaluates the portfolio against defined risk thresholds (sector concentration, single-holding concentration, low diversification) and lists an actionable item for each threshold crossed, each with a link to the relevant page or section.

Alternative flows:

- Nothing to flag: no thresholds are crossed; the Action Centre displays a reassuring all-clear state instead of an empty card.
- No holdings: the health score and Action Centre display an onboarding prompt to import a portfolio.

TUCEW the user has seen their health score and knows what, if anything, needs their attention.

4. Functional Requirements

Functional requirements are stated in terms of what the user can do and what the system must visibly provide.

4.1 Subsystem 1: Authentication

FR-1: User Registration

- FR-1.1: The system shall allow a new user to register with their full name, email address and password.
- FR-1.2: The system shall validate that the email address is correctly formatted before the registration is submitted.
- FR-1.3: The system shall enforce a password policy of at least 8 characters, containing at least one uppercase letter, one lowercase letter, one number and one special character.
- FR-1.4: The system shall display a clear inline error message next to each invalid or missing registration field.
- FR-1.5: The system shall reject a registration that uses an email address already associated with an existing account, and tell the user to sign in instead.

FR-2: Email Verification

- FR-2.1: The system shall send a 6-digit verification code to the user's email address upon registration.
- FR-2.2: The system shall require the user to enter the verification code before the account can be used.
- FR-2.3: The system shall reject invalid or expired codes with a clear error message and allow the user to retry.
- FR-2.4: The system shall direct the user to the login page once their email is verified.

FR-3: Login and Session

- FR-3.1: The system shall authenticate users by email address and password.
- FR-3.2: The system shall display an appropriate error message when invalid credentials are provided, without revealing whether the email or the password was incorrect.
- FR-3.3: The system shall keep the user's session active while they work, without requiring re-authentication on every page.
- FR-3.4: The system shall redirect an unauthenticated user who attempts to open a protected page to the login page.
- FR-3.5: The system shall return an authenticated user who opens the login or registration page to their dashboard.
- FR-3.6: The system shall allow the user to log out; after logging out, protected pages shall no longer be accessible until the user logs in again.

FR-4: Multi-Factor Authentication

- FR-4.1: The system shall require a second authentication factor, a 6-digit time-based code from an authenticator app, in addition to the password.
- FR-4.2: On first login, the system shall guide the user through pairing an authenticator app by scanning a QR code or entering a setup key manually.
- FR-4.3: The system shall require the user to confirm the pairing by entering a code generated by their authenticator app before MFA is activated.
- FR-4.4: On subsequent logins, the system shall prompt for the current authenticator code after the password is accepted.
- FR-4.5: The system shall reject an incorrect authenticator code with a clear error message and allow the user to retry.

4.2 Subsystem 2: Portfolio Management

FR-5: File-Based Portfolio Import

- FR-5.1: The system shall allow the user to import portfolio data by uploading a CSV file.
- FR-5.2: The system shall allow the user to import portfolio data by uploading an Excel workbook that follows the downloadable template provided by the system.
- FR-5.3: The system shall allow the user to import portfolio data by uploading a brokerage statement, from which holdings, transactions, dividends and statement metadata are extracted automatically.
- FR-5.4: The system shall provide downloadable import templates (Excel) so the user knows the expected format.
- FR-5.5: The system shall validate every uploaded file and reject unsupported or malformed files with a clear error message that allows the user to retry.
- FR-5.6: The system shall display a preview of the parsed holdings for the user to confirm before anything is saved.
- FR-5.7: The system shall inform the user when a file parses successfully but contains no recognisable holdings, and prompt them to enter data manually or try a different file.

FR-6: Manual Data Management

- FR-6.1: The system shall allow the user to manually add a holding by capturing its ticker symbol, quantity and purchase price.
- FR-6.2: The system shall validate manually entered ticker symbols against known listed stocks before saving.

FR-7: Holdings Ownership and Display

- FR-7.1: The system shall associate every imported or manually entered holding with the account of the user who created it.
- FR-7.2: The system shall never display one user's portfolio data to any other user.

- FR-7.3: The system shall display the user's holdings in a structured table showing ticker symbol, company name, sector, quantity, average purchase price, current price, total value, profit/loss in Rand, and profit/loss as a percentage.
- FR-7.4: The system shall visually distinguish gains from losses (gains in green and losses in red).

4.3 Subsystem 3: Analytics and Financial Indicators

FR-8: Financial Indicator Calculation

The system shall calculate the following seven indicators from the user's actual holdings and market data. Each was selected for its value to a retail investor, as motivated in the indicator tooltips.

- FR-8.1: The system shall calculate and display the Sharpe Ratio (risk-adjusted return: portfolio return minus risk-free rate, divided by portfolio standard deviation).
- FR-8.2: The system shall calculate and display the Sortino Ratio (as the Sharpe Ratio, but penalising only downside deviation).
- FR-8.3: The system shall calculate and display the P/E Ratio for individual holdings.
- FR-8.4: The system shall calculate and display the CAPM expected return for individual holdings.
- FR-8.5: The system shall calculate and display the Altman Z-Score for individual holdings, classifying results as safe (above 2.99), grey zone (1.81 to 2.99) or distress (below 1.81).
- FR-8.6: The system shall calculate and display Beta for individual holdings relative to the JSE All Share Index.
- FR-8.7: The system shall calculate and display the 14-day RSI, classifying results as overbought (above 70), neutral (30 to 70) or oversold (below 30).

FR-9: Indicator Presentation

- FR-9.1: The system shall provide a plain-language explanation for each indicator, available on hover or keyboard focus, describing what the indicator means and how to interpret its value.
- FR-9.2: The system shall display an N/A state with a short explanation when there is insufficient data to calculate an indicator, without affecting the other indicators.
- FR-9.3: The system shall colour-code each indicator value as positive, neutral or negative according to the classification thresholds defined in FR-8.
- FR-9.4: The system shall allow the user to retry a failed calculation from the affected indicator card while the rest of the page remains usable.

4.4 Subsystem 4: Market Data

FR-10: Market Data Provision

- FR-10.1: The system shall retrieve current prices, daily change percentages and trading volumes for every holding in the user's portfolio.

- FR-10.2: The system shall retrieve historical price data over selectable periods for charting and indicator calculation.
- FR-10.3: The system shall allow the user to search for a stock by ticker symbol or company name and see matching results.
- FR-10.4: The system shall continue functioning when live market data for a holding is unavailable, showing the most recent known values or a clear unavailable state for that holding only.

4.5 Subsystem 5: News Correlation

FR-11: Price Movement Detection

- FR-11.1: The system shall monitor daily price movements for every holding in the user's portfolio.
- FR-11.2: The system shall flag a movement as significant when the day's price change exceeds a configured threshold (default: 5% in either direction).

FR-12: News Retrieval and Correlation

- FR-12.1: The system shall retrieve financial news articles from its news data providers published within a 48-hour window of a detected significant price movement.
- FR-12.2: The system shall show only articles relevant to the specific holding's ticker symbol or company name.
- FR-12.3: The system shall display each correlated article with its headline, source, publication time, related ticker badge, and an indicator of how the related stock's price moved.
- FR-12.4: The system shall allow the user to filter displayed articles by ticker symbol.
- FR-12.5: The system shall allow the user to filter displayed articles by date range.
- FR-12.6: The system shall clearly indicate when no news is available for a holding or filter combination, and prompt the user to broaden their filters.

4.6 Subsystem 6: AI Assistant

FR-13: Conversational Interface

- FR-13.1: The system shall provide a chat interface where the user can submit natural-language questions about their portfolio.
- FR-13.2: The system shall ground every response in the user's own data by combining their holdings, calculated indicators and relevant market data with the question before it reaches the AI model.
- FR-13.3: The system shall display suggested prompts when the chat is empty, which the user can send with a single click.
- FR-13.4: The system shall display an error message and allow retry when a response cannot be generated.

FR-14: Conversation Management

- FR-14.1: The system shall save the user's conversations so they persist across sessions.
- FR-14.2: The system shall display the user's saved conversations in a list, and load a conversation's full message history when selected.
- FR-14.3: The system shall allow the user to start a new conversation at any time.
- FR-14.4: The system shall allow the user to rename a saved conversation.
- FR-14.5: The system shall allow the user to delete a saved conversation.

FR-15: Compliance and Guardrails

- FR-15.1: The system shall never provide explicit buy or sell recommendations, regardless of how a question is phrased.
- FR-15.2: When the user asks for explicit investment advice, the system shall respond with a disclaimer explaining that EquityLens provides analysis and information only, and direct the user to the Analytics page.
- FR-15.3: The system shall display a disclaimer with AI responses clarifying that they are informational and not regulated financial advice.

4.7 Subsystem 7: Dashboard and Insights

FR-16: Portfolio Overview

- FR-16.1: The system shall display the portfolio's total value, total gain/loss (Rand and percentage), and today's change.
- FR-16.2: The system shall summarise, in one plain-language sentence, how the portfolio moved today and which holding drove the move.
- FR-16.3: The system shall display the user's largest positions with each position's share of the total portfolio and its daily change.
- FR-16.4: The system shall display the portfolio's sector allocation, and allow the user to select a sector to see the holdings within it.

FR-17: Performance vs Benchmark

- FR-17.1: The system shall record one snapshot per day of the portfolio's total value and the JSE All Share Index level.
- FR-17.2: The system shall chart portfolio value and benchmark together over time, and display the portfolio return, benchmark return, and the difference for the charted window.
- FR-17.3: The system shall leave a visible gap in the benchmark series for days on which the benchmark could not be fetched, rather than substituting a fabricated value.
- FR-17.4: The system shall display monthly dividend income derived from the user's imported dividend records.

FR-18: Portfolio Health and Action Centre

- FR-18.1: The system shall compute a portfolio health score out of 10 from the user's actual holdings, combining diversification, concentration risk, today's momentum, and performance relative to the benchmark.
- FR-18.2: The system shall display the health score with a classification (Excellent, Healthy, Mixed, Needs attention) and identify the strongest and weakest contributing factors in plain language.
- FR-18.3: The system shall flag an actionable item when a defined risk threshold is crossed: a single sector at 40% or more of the portfolio, a single holding at 30% or more, or fewer than 5 positions held.
- FR-18.4: Each actionable item shall state the risk in plain language and link the user to the page or section where they can act on it.
- FR-18.5: The system shall show a reassuring all-clear state when no thresholds are crossed, and an onboarding prompt when the user has no holdings.

FR-19: General Interface

- FR-19.1: The system shall allow the user to switch between light and dark display themes, and remember the choice.
- FR-19.2: The system shall display reference market context (JSE All Share Index level, USD/ZAR exchange rate, Brent crude price).
- FR-19.3: The system shall provide an in-product Help menu giving the user access to guidance, tutorials and frequently asked questions.

4.8 Subsystem 8: Watchlist

FR-20: Watchlist Management

- FR-20.1: The system shall allow the user to add a stock they do not own to their watchlist by searching for it by ticker or company name.
- FR-20.2: The system shall display each watchlist entry with its ticker, company name, sector, current price and daily percentage change.
- FR-20.3: The system shall allow the user to remove an entry from their watchlist.
- FR-20.4: The system shall keep each user's watchlist private to that user.

5. Non-Functional Requirements

5.1 NFR-1: Security

- NFR-1.1: 100% of API endpoints, other than the health check and the registration/login/verification endpoints, shall reject unauthenticated requests with an HTTP 401 response.
- NFR-1.2: User credentials shall never be stored in plain text by the system. Sessions shall be established with expiring access tokens with a lifetime of at most 60 minutes; an expired session shall require re-authentication or a silent token refresh before any protected resource is served.
- NFR-1.3: Multi-factor authentication (TOTP) shall be enforced on user accounts.
- NFR-1.4: All data in transit shall be encrypted with TLS; all data at rest shall be encrypted with AES-256.
- NFR-1.5: All portfolio data shall be scoped to the owning user at the data-access level: 0 instances of cross-user data exposure, verified by integration tests.
- NFR-1.6: The system shall never store raw brokerage credentials; portfolio data enters the system only by file upload or manual entry.
- NFR-1.7: All personal data handling shall comply with POPIA.

5.2 NFR-2: Performance

- NFR-2.1: API endpoints shall respond within 500 ms at the 85th percentile under normal load.
- NFR-2.2: Initial page load shall complete within 5 seconds on a standard broadband connection.
- NFR-2.3: Dashboard visualisations shall render within 3 seconds for portfolios of up to 50 holdings.
- NFR-2.4: Market data shall be served through a cache-aside store: a request for an instrument that is already cached shall be answered with zero calls to the upstream provider, keeping repeat dashboard loads within provider rate limits.

5.3 NFR-3: Reliability

- NFR-3.1: The failure of any single external data source (market data, news, AI model) shall degrade only its own feature; the remainder of the system shall stay fully functional.
- NFR-3.2: If the primary AI service fails, the fallback shall activate within 2 seconds with no user-visible error.
- NFR-3.3: A failed fetch for one holding's market data shall not prevent the dashboard from rendering the remaining holdings.

5.4 NFR-4: Scalability

- NFR-4.1: The backend shall support at least 50 concurrent users without performance degradation beyond the NFR-2 targets.

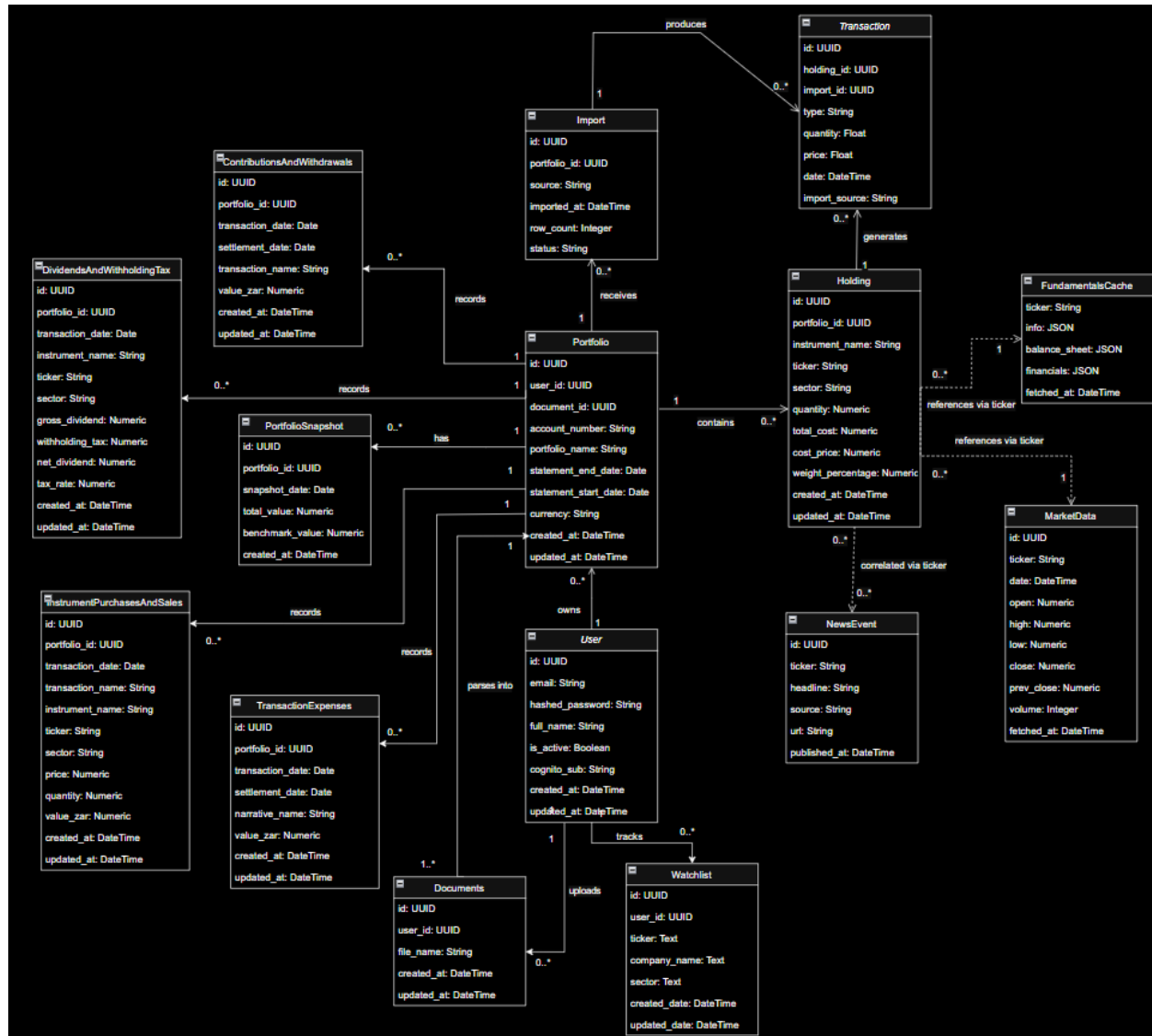
- NFR-4.2: Database query times shall remain below 200 ms at the 95th percentile with up to 1 million rows of historical market data.
- NFR-4.3: The system shall accommodate a 200% workload increase without major architectural changes and with no more than a 10% performance decrease.

5.5 NFR-5: Usability and Accessibility

- NFR-5.1: A new user shall be able to register, import a portfolio and reach a populated dashboard within 5 minutes of first use.
- NFR-5.2: The interface shall meet WCAG 2.1 AA: all interactive elements keyboard-navigable with visible focus states, and text contrast ratios of at least 4.5:1.
- NFR-5.3: Every financial indicator shall have a plain-language explanation accessible from where the indicator is displayed, so no finance training is required to use the system.
- NFR-5.4: The interface shall be fully responsive across desktop (1280px+), tablet (768-1279px) and mobile (320-767px) with no horizontal scrolling on mobile.
- NFR-5.5: The interface shall respect the user's reduced-motion preference and disable non-essential animations when set.

6. Domain Model

6.1 UML Class Diagram



6.2 Entity Descriptions

User - an account holder. `cognito_sub` links the local record to the identity provider; no password material is stored locally for identity-provider accounts. Each user's data (documents, portfolios, watchlist, conversations) is reachable only through their own account.

Document - a record of an uploaded statement file (PDF/CSV/Excel). Retained so an import can be traced back to its source file.

Portfolio - a set of holdings imported from one statement or created manually, including the statement period it covers. A user may hold multiple portfolios.

Holding - a single position within a portfolio: the instrument, quantity held, and cost basis (total cost and per-share cost price). Current value is derived at read time from market data, not stored.

InstrumentTransaction - a buy or sell recorded in the statement, with date, price, quantity and Rand value.

ContributionWithdrawal - cash paid into or withdrawn from the account.

DividendRecord - a dividend event including gross amount, withholding tax and net amount, used for the dividend income view.

TransactionExpense - fees and costs recorded in the statement.

PortfolioSnapshot - one row per portfolio per day recording total portfolio value and the JSE All Share Index level, accumulated to build the performance-vs-benchmark history. Uniqueness is enforced on (portfolio, date).

WatchlistEntry - a stock the user tracks without owning. Private to the user.

Conversation - a saved AI assistant conversation with a user-editable title.

Message - a single message within a conversation; role distinguishes the user's messages from the assistant's.

MarketData (cache) - fetched price data (price, volume, daily change, fetch timestamp) retained to serve repeat requests within the cache window. It is a cache, not a system of record.

NewsEvent (cache) - a retrieved news article (headline, source, URL, publication time, related ticker) retained for display and correlation

7. Requirements-Use Case Traceability

[illegible]

FR-15				X							
FR-16		X									X
FR-17		X								X	
FR-18		X									X
FR-19		X									
FR-20								X			

8. Assumptions and Dependencies

8.1 Assumptions

- A-1: Users have a basic understanding of stock ticker symbols and financial terminology.
- A-2: Users have an existing investment portfolio with at least one holding, or intend to build one.
- A-3: Users have a modern web browser with JavaScript enabled, and access to an authenticator app on a phone or desktop for MFA.
- A-4: Portfolio PDF statements are sourced from supported South African brokerages, or follow the provided import templates.
- A-5: Market data supplied by the market data provider accurately reflects JSE-listed stock prices.

8.2 Dependencies

- D-1: Primary market data provider. If unavailable: the fallback provider activates; if both fail, the last cached values are shown.
- D-2: News data providers. If unavailable: the news correlation feature is unavailable; the rest of the system is unaffected.
- D-3: AI model provider. If unavailable: the assistant falls back to the secondary layer; if both fail, an error prompts the user to retry.
- D-4: Cloud free-tier allowance. If exceeded: infrastructure costs would be incurred, violating the project's cost constraint.
- D-5: CI pipeline provider. If unavailable: automated quality gates are unavailable and manual testing is required before merges.
- D-6: PDF text extraction capability. If unavailable: PDF import is unavailable; CSV, Excel and manual entry remain functional.